
Caleb Chia

Aspiring physicist, current national serviceman

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About

Physics student with multiple olympiad achievements across physics, nuclear science, chemistry, mathematics, and informatics. Experienced with academic research and technical leadership. Currently serving 2-year national service in the Singapore Armed Forces as an army intelligence specialist.

Education

St. Edmund's College, University of Cambridge BA (Hons) Natural Sciences (Physical), commencing October 2027	2027–2030
Anglo-Chinese School (Independent), Singapore International Baccalaureate Diploma Programme (Nov 2024): 43/45 Graduated: December 2024	2019–2024

Academic Activities

Physics and Nuclear Science Olympiads Selected for the Singapore Physics Olympiad Training team. Trained with the National University of Singapore and Singapore Nuclear Research and Safety Institute. Represented Singapore at the 2024 International Nuclear Science Olympiad and won a Gold medal.	2024
Engineering Intern, BorgWarner Short work attachment in the electrical engineering department. Worked on circuit design, SPICE-based simulations, RF-testing and bench-testing of devices (mainly high-power inverters).	2024
Research Intern, Nanyang Technological University Conducted a year-long research project on GaN HEMTs, characterizing and analyzing device behavior to optimize for high-frequency performance. Final report titled: " <i>Surface Passivation with Al₂O₃/SiN Bilayer in Advanced Gallium Nitride High Electron Mobility Transistors (HEMTs)</i> ," defended 29 February, 2024.	2023–2024
Executive Committee, ACS(I) Coding Competition Team Organized and helped develop <i>Hack@AC 2024</i> , a national pre-university CTF. Conducted training sessions and represented the school in cybersecurity and competitive programming competitions.	2023–2024
Vice-President, ACS(I) Mathematics Competition Team Led internal competitions, mentored juniors and represented school in math competitions.	2022–2024

Non-Academic Activities

Army Intelligence Specialist, Singapore Armed Forces Trained in section-level military leadership, ground reconnaissance, intelligence staffwork and analysis.	2025–Present
Audio-Visual Director, Bukit Panjang Methodist Church Lead AV technician for youth worship hall. Mentored junior volunteers, managed scheduling, training and system upgrades.	2024–Present
Class Chairperson, ACS(I) Planned and organized class activities. Awarded ACS(I) Pastoral Care Award.	2024
Volunteer Chess Teacher, Project Reperio Taught chess to Rohingya students as part of a holistic education programme planned by <i>Project Reperio</i> , in collaboration with <i>Cahaya Surya Bakti</i> .	2023–2024
Volunteer Tutor, Heartfriends Student Care Centre Managed volunteer scheduling, tutored primary students in math and engaged them with games, as part of a 4-month class project.	2023

Honors & Awards

- Gold Medal, International Nuclear Science Olympiad (2024)
- Gold Medal, Singapore Physics Olympiad (2023)
- Silver Medal, National Olympiad for Informatics (2023)
- Silver Medal, Singapore Chemistry Olympiad (2023)
- Silver Medal, Singapore Mathematical Olympiad (2022, 2023, 2024)
- Distinction (Most Comprehensive Analysis), Singapore International Mathematical and Computational Challenge (2024)
- Distinction, International Mathematical Modelling Challenge (2024)
- Gold Award, Nanyang Research Programme (2024)
- 3rd Team Placing, Cyberthon (2024)
- High School Dean's List - Physics, Chemistry, Mathematics (2024)
- MOE Engineering and Tech Programme Scholarship (2023)
- ABRSM Piano Grade 8 Certification (2024)

Standardized Testing

IBDP: 43/45 HL Physics (7), HL Chemistry (7), HL Mathematics AA (7), SL English A: Lang & Lit (6), SL Economics (7), SL Chinese B (7)	November 2024
SAT: 1570/1600 Math: 800, EBRW: 770	October 2024

Other Interests

Running, cycling, movies, reading